



Students' Personal Norms and Intentions to Reduce Food Waste in Universities: Moderating Role of the Food Purchasing Methods



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Abstract: University students, as a key youth consumer demographic, will play a vital role in shaping sustainable purchasing behavior in the future. This study aims to uncover the factors influencing students' intention to minimize food waste at universities using the extended norm activation theory. An online survey of 664 students examined intentions to reduce food waste on campus. Of these, 245 students used online food delivery (OFD), while 419 engaged in in-canteen dining (IC). To evaluate the empirical data, this study utilized a partial least squares structural equation model and executed measurement invariance testing within the composite model. The empirical results demonstrate that the activation of personal norms is driven by awareness of consequence and the ascription of responsibility, which consequently has a direct impact on the intention to reduce food waste. Personal norms also indirectly influence the intention to minimize food waste. Students who purchased meals only reported weaker personal norms and lower intention to reduce food waste than those who ate in the canteen. However, the OFD group showed greater awareness of consequence, which supported their efforts to reduce food waste, compared with the IC group. Overall, this study provides further insight into the psychological mechanisms underlying sustainable food consumption among university students.

Keywords: Intention to reduce food waste; Personal norms; Food purchasing methods; Online food delivery; In-canteen dining

1. Introduction

Food waste is a critical issue because it threatens food security (Wang et al., 2022b), exacerbates climate change and greenhouse gas emissions (Kallbekken & Sælen, 2013; Queded et al., 2013), and incurs economic losses (Hennchen, 2019). This issue is particularly acute in sectors such as hospitality, tourism, and restaurants (Okumus, 2020). A growing body of research examines food waste habits in contexts prone to overconsumption, including restaurants, hotels, and households. Nevertheless, the specific factors driving food waste behavior among distinct demographic segments, such as university students, remain under-explored (Lazell, 2016). Universities are recognized as an important area of concern due to increasing food waste (Yui & Biltekoff, 2021). However, higher education institutions, school food service managers, and especially students have shown minimal interest in reducing food waste. Universities with substantial student enrollments manage a considerable number of meals in a single location (Wilkie et al., 2015). At the same time, university students aged 18–24, often living independently for the first time, make choices about food and other consumer products without the guidance they may have received at home. This group often lacks time to prepare meals at home or seeks to minimize the effort involved in cooking and cleaning. Irregular eating patterns, limited knowledge of food preservation, and time constraints contribute substantially to food waste among students (Ozanne et al., 2022). Furthermore, existing literature demonstrates that students frequently encounter difficulties regarding food management, which potentially results in surplus food handling (Miśniakiewicz et al., 2024). Research among university students indicates that annual food waste per student is approximately 142 pounds (Wang et al., 2022a). University students' food waste habits differ from those of households. Consumers interact with food primarily in its consumable form. Therefore, the causes of food waste largely stem from factors such as overordering, inability to preserve leftovers, and failure to

meet quality standards (Farr-Wharton et al., 2014). As members of a young consumer demographic, university students may significantly influence future consumption patterns by establishing behaviors during their academic years (Tsai et al., 2020).

Furthermore, prior studies have demonstrated on explaining food waste intentions and behaviors through individual motivations across common demographics, including tourists, Generation Z, and household consumers (Lorenz-Walther et al., 2019). From a societal perspective, minimizing food waste is viewed as an environmentally beneficial action. However, limited research has examined how social motivations influence consumer behavior (Kaiser et al., 2005). This gap is especially relevant in universities, where students, given their similar education, knowledge, and skills, are expected to uphold higher ethical and environmental standards than consumer groups in other sectors. Amid substantial shifts in food consumption, particularly during the Covid-19 pandemic, online food delivery (OFD) services have transformed university students' eating habits. In addition to in-canteen dining (IC), students can now order meals through OFD services. OFD is increasingly adopted by young people worldwide and may shape students' food purchasing and consumption patterns, thereby influencing food waste behavior (Wang et al., 2024; Wei et al., 2023). OFD services provide a business model that supports ordering, payment, and process monitoring, although they do not prepare food. Through mobile applications and other internet-based platforms, OFD systems offer a range of food options, process orders, communicate them to food providers, manage payment, coordinate delivery, and provide tracking functions. OFD companies have also invested heavily in promotions and campaigns, subsidizing participating restaurants and offering consumer meals at reduced prices or through other incentives (Pigatto et al., 2017; Zaheer et al., 2024). Although OFD is regarded as a technological advancement in the food service sector, it has been criticized for intensifying food waste in several countries (Li et al., 2020). Therefore, future studies should focus on exploring the impact of food procurement and consumption habits on the food wasting tendencies of younger demographics.

Several prior studies have used distinct theories, including Social Cognitive Theory (Deavin et al., 2018), Social Practice Theory (Yui & Biltekoff, 2021), the Theory of Planned Behavior (TPB) (Wu et al., 2019), and Norm Activation Theory (NAT) (Wang et al., 2022b), to assess consumers' intentions to reduce food waste in restaurant contexts. This study argues that students' food waste reduction behavior is driven by environmental motivations and identifies NAT as the most suitable framework for explaining this demographic's intention and actions. Under the NAT framework, students' intentions to mitigate food waste are conceptualized as being driven by personal norms, which are triggered by key antecedents such as awareness of consequence and ascription of responsibility. The study also considers the growth of OFD platforms and examines the moderating role of OFD purchasing and canteen dining on the relationships between the components of the NAT and the intention to reduce food waste. Finally, this study endeavors to address theoretical and practical shortcomings in food waste research, particularly the gap in understanding the psychological determinants of student behavior at universities.

2. Theoretical Framework and Research Hypotheses

2.1 Norm Activation Theory

NAT is a theoretical framework for predicting an individual's behavioral intentions toward prosocial behavior (Schwartz, 1977). Accordingly, NAT suggests that the strength of an individual's moral obligation influences their support for the behavior. Applying NAT to pro-environmental conduct is appropriate, as pro-environmental actions are often viewed as a subset of pro-social behavior because of their beneficial outcomes (Setiawan et al., 2020). NAT posits that personal norms develop when individuals recognize the consequences of not performing a given action and accept responsibility for the potential negative outcomes. Once established, these personal norms motivate appropriate behavior. This study argues that food waste reduction is environmentally beneficial because it primarily serves ecological values rather than providing direct benefits to the individual, particularly by reducing waste from excess food. It therefore reflects behavior intended to benefit others without providing any direct personal gain (De Groot & Steg, 2009).

2.2 Awareness of Food Waste Consequences

Awareness of consequence is defined as a person's recognition of the negative outcomes for others or for other values when a given conduct is not undertaken (Schwartz & Boehnke, 2004). The NAT model posits that self-attributed responsibility activates personal norms, which in turn encourage or discourage individuals from engaging in behavior that reduces negative outcomes (Gao et al., 2017). Food waste is widely regarded as highly harmful to the environment; consequently, awareness of its consequences helps individuals recognize its adverse effects, thereby shaping ethical standards and fostering a commitment to waste-reduction practices. Understanding these consequences is essential for shaping personal standards for reducing food waste (Ellison et al., 2019; Fraj-Andrés et al., 2022). When consumers reflect on the adverse effects of food waste, including significant environmental and societal challenges, they may develop strategies to mitigate these consequences, thereby

fostering individual food-saving standards (Watson & Meah, 2012). Teng et al. (2022) found that personal norms and awareness of consequence are significant determinants of reduced food waste intention in the restaurant and hotel sector. Panda et al. (2024) contend that sustainable behavior in household food consumption is nurtured by an awareness of the environmental and economic ramifications of food waste. Therefore, we propose the following hypotheses:

H₁. Awareness of consequence positively influences personal norms of students at universities.

H₂. Awareness of consequence positively influences students' intentions to reduce food waste at universities.

2.3 Ascription of Responsibility for Food Waste

Ascription of responsibility refers to an individual's perceived responsibility for the negative consequences arising from failing to engage in a societal behavior (De Groot & Steg, 2009). Ascription of responsibility denotes the extent to which an individual assesses their accountability for a specific behavior (Schwartz, 1977). He & Zhan (2018) suggested that responsibility is ascribed when consumers perceive themselves as accountable for the harmful impacts on the environment caused by their purchasing habits. It relates to personal actions that may reduce or exacerbate negative effects (Stern, 2000) resulting from the failure to engage in socially beneficial behaviors (Talwar et al., 2022). In this context, the activation of personal norms occurs once individuals acknowledge their accountability for the adverse outcomes associated with food waste. Reducing food waste is a collective societal obligation, with each individual accountable for its adverse effects (Programme, 2024). This sense of responsibility fosters constructive personal norms for reducing food waste. Consequently, when individuals view responsibility as intrinsic, they are more likely to engage in mitigation behavior (Wang et al., 2024). Research shows that individuals who perceive a personal responsibility for environmental issues are more likely to engage in pro-social behaviors (Ma et al., 2023; Le et al., 2024). Accordingly, we propose the following hypotheses:

H₃. Ascription of responsibility positively influences personal norms of students at universities.

H₄. Ascription of responsibility positively influences students' intentions to reduce food waste at universities.

2.4 Personal Norms

Personal norm is described as an individual's awareness of their moral obligation that either encourages or inhibits specific behavior (Schwartz & Boehnke, 2004). This suggests that personal norms serve as ethical motivators embedded in behavior and decision-making, linking conscience with action. Moreover, they serve as internal motivations that shape attitudes and responses to perceived social or environmental concerns (Wang et al., 2022b). Personal norms are a key concept in behavioral psychology. The contemplation of the adverse outcomes of one's actions on others often fosters a heightened sense of moral obligation, which subsequently drives intentions toward socially beneficial behaviors (Hong et al., 2024). This study posits that minimizing food waste constitutes a code of conduct and an ethical imperative, thereby inducing feelings of shame or guilt among those who waste food (Djekic et al., 2019). Accordingly, when individuals activate personal norms, they engage in waste-reducing actions that can generate positive emotions (Onwezen et al., 2013). Setiawan & Puspitasari (2023) examined the impact of personal ethical standards on food waste reduction behavior in restaurants, concluding that moral obligation is a significant predictor of environmentally responsible behavior. Panda et al. (2024) investigated home food management, finding that personal norms significantly influence decisions to reuse or discard food. Nevertheless, other perspectives assert that personal norms do not consistently encourage sustainable consumption behavior. Consumers may hold strong ethical standards, yet consumption-stimulating policies and programs (such as discounts and gifts) can complicate efforts to reduce food waste (Fraj-Andrés et al., 2022). Consequently, we propose the following hypotheses:

H₅. Personal norms positively influence students' intentions to reduce food waste at universities.

H₆. Personal norms mediate the relationship between consequence awareness and the intention to reduce food waste among students at universities.

H₇. Personal norms mediate the relationship between responsibility awareness and the intention to reduce food waste of students at universities.

All the above-mentioned relationships and their corresponding hypotheses are illustrated in the following conceptual model (Figure 1).

2.5 Food Purchasing Methods: Online Food Delivery vs. In-Canteen Dining

This study categorizes students into two groups: (1) individuals who regularly dine at the canteen (IC) and (2) individuals who frequently use OFD services (Wang et al., 2022a). The OFD group orders meals from local food retailers and restaurants via mobile applications developed by third-party providers such as Grab and Uber, or by the establishments themselves, such as Domino's and McDonald's (Wang et al., 2024). The canteen group

comprises consumers who eat at the establishments located on the university campus. Recent research indicates that both canteen dining and OFD are directly associated with food waste behavior. OFD platforms frequently involve business alliances that set minimum order thresholds. This mechanism encourages consumers to purchase additional items or engage in group buying to obtain better prices, secure free delivery, or qualify for promotions. Consequently, these behaviors can lead to excessive food purchases and an increased risk of food waste (Wei et al., 2023). Under these circumstances, consumers' awareness of consequence, ascription of responsibility, and personal norms are critical determinants that influence food selection, purchasing intentions, and food waste reduction behaviors.

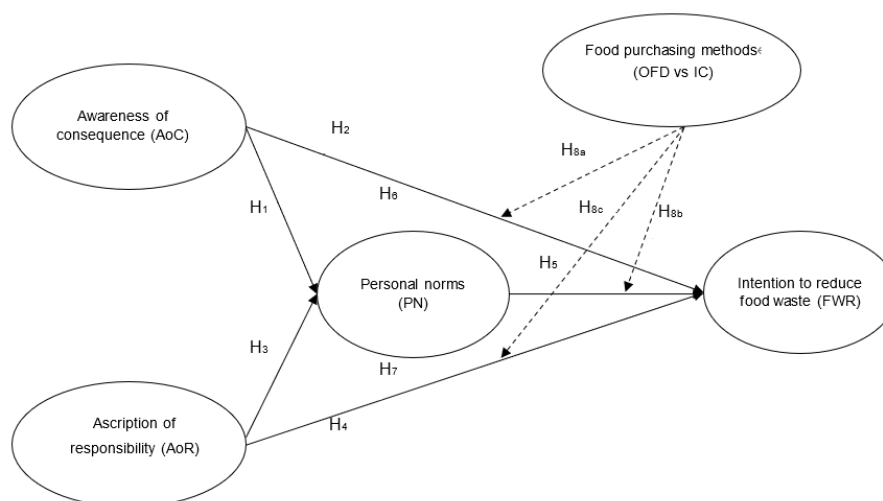


Figure 1. Proposed research model

Conversely, purchasing food at a canteen allows consumers to directly perceive and evaluate items through sensory cues related to taste, portion size, and appearance. In addition, canteen environments typically induce vendor recommendations and visual prompts or slogans that encourage mindful consumption. These physical and social cues help consumers more accurately assess their needs and regulate purchasing behavior, making them less susceptible to cognitive influences such as awareness of consequence, ascription of responsibility, or subjective norms (Li et al., 2020). Consequently, the following hypotheses were formulated:

H₈. Food purchasing methods (OFD and IC) moderate the relationship between awareness of consequence (H_{8a}), personal norms (H_{8b}), awareness of responsibility (H_{8c}) and students' intention to reduce food waste at university.

3. Methodology

3.1 Sample and Data Collection

To assess the interrelationships among the variables outlined in the conceptual framework, a comprehensive survey was administered via QR codes scanned through the Zalo application at higher education institutions across major Vietnamese cities. We conducted a pilot survey in January 2025, evaluated the results and made revisions, and then conducted the main survey from March to April 2025. We selected students as participants because they represent future consumers who are likely to shape consumption patterns in the coming decades. Moreover, they are in the process of forming their personal identity and developing belief and value systems. Because habits formed during this period often persist into adulthood, this group offers policymakers a valuable opportunity to promote sustainable consumption practices in Vietnam. In addition, as the next generation of adult consumers, students have considerable purchasing power and can influence food choices and consumption patterns within their households. We also chose students because, as young people with educational qualifications, they are more likely to understand environmentally sustainable consumer behavior (Vermeir & Verbeke, 2008). After removing incomplete and outlier surveys, we obtained 664 usable responses for data analysis, yielding a response rate of 73.5%.

3.2 Scales

Prior to administering the survey, we pre-tested it by interviewing a focus group of 10 students who represented the target respondents to assess construct validity (Sekaran, 2016). The questionnaire used for the formal study

was organized into three sections. The first section collected information on students' food procurement and consumption behaviors at the university, including frequency, purchasing methods, meal types (OFD or IC), expenses, and related factors. To evaluate participants' food waste reduction intentions, the second section utilized a 5-point Likert scale, with response options anchoring from 'strongly disagree' (1) to 'strongly agree' (5). The final section captured students' demographic characteristics.

3.3 Methods of Data Analysis

Partial least squares structural equation modeling (PLS-SEM) was adopted as the core methodology for this research, primarily owing to its robustness in handling intricate models with numerous latent variables and data non-normality (Hair et al., 2019). We selected PLS-SEM rather than covariance-based SEM for three reasons: (i) the proposed model incorporates both moderating and mediating relationships; (ii) it enables the estimation and evaluation of structural paths within the research model (Hair et al., 2021). Analyses were conducted in SmartPLS4 following the two-step recommendation procedure (Hair et al., 2019). To assess whether structural relationships differed between two purchasing models (OFD vs. IC), we performed PLS multigroup analysis, in which between-group path differences represent effect sizes (Sarstedt et al., 2011). Because meaningful multigroup comparisons require measurement invariance, we evaluated invariance across groups prior to hypothesis testing using the measurement invariance of composites (MICOM) procedure, which tests (1) configural invariance, (2) compositional invariance, and (3) equality of composite means and variances via permutation (Hair et al., 2019).

4. Results

4.1 Descriptive Analysis

Among the 664 students surveyed, 371 were female (55.87%) and 293 were male (44.13%). First-year students accounted for 34.79% of the sample, followed by second-year (29.37%), third-year (25.15%), and fourth-year (10.69%). In this survey, 48.04% of students reported dining on campus 1–2 times per week, 37.35% reported dining on campus 3–4 times per week, and 14.61% reported dining on campus more than 5 times per week. 93.22% of students reported eating lunch, 60.39% reported eating breakfast, 29.07% reported eating dinner, and 43.07% reported eating snacks.

Regarding purchasing methods, 63.10% of students reported dining at the campus canteen, while 36.90% reported using OFD. Regarding living arrangements, 67.92% of respondents reported living in rented accommodations, while 32.08% reported living with family or relatives. Regarding meal expenditures, 22.14% of students reported spending less than 30,000 VND on their most recent lunch, 46.84% reported spending 30,000 VND–50,000 VND, 26.51% reported spending 50,000 VND–100,000 VND, and 4.52% reported spending more than VND 100,000. Finally, 30.42% of students reported discarding less than 10% of the food from their most recent lunch at university. In contrast, 48.64% reported leaving 10%–30% uneaten, 16.11% reported wasting 30%–50%, and 4.82% reported consuming less than 50% of the food they ordered.

4.2 Measurement Framework

As presented in Table 1, the reliability of all constructs was established, with Cronbach's alpha coefficients spanning from 0.779 to 0.835. To evaluate convergent validity, a triad of benchmarks was employed: outer loadings above 0.70 (Choi & Chung, 2013; Hair et al., 2014); an average variance extracted (AVE) exceeding 0.5 (Hair et al., 2019); and the composite reliability surpassing 0.70 (Fornell & Larcker, 1981; Hair et al., 2019; Wang et al., 2022b). The empirical data met all thresholds, exhibiting a minimum outer loading of 0.764, AVE values between 0.633 and 0.736, and CR estimates ranging from 0.780 to 0.836, thereby verifying satisfactory convergent validity.

Table 1. Reliability and convergent validity

Variables	Code	Outer Loading	Cronbach's Alpha	CR	AVE
Ascription of responsibility (AoR) (De Groot & Steg, 2009; Kim et al., 2022)	AoR ₁ As a student, I feel responsible for reducing food waste during my meals at the university.	0.823	0.806	0.807	0.633
	AoR ₂ As a student, I feel responsible for the adverse environmental impacts of food waste at my university.	0.805			
	AoR ₃ As a student, I feel remorse about leaving leftovers from my meals at my university.	0.789			

Awareness of consequence (AoC) (De Groot & Steg, 2009; Kim et al., 2022)	AoR ₄ Students have greater responsibility for the environmental pollution and ecological degradation resulting from food waste in the university.	0.764			
	AoC ₁ Food waste in universities constitutes a significant environmental issue.	0.815			
	AoC ₂ Food waste from university meals increases carbon dioxide emissions and adds to climate change.	0.859	0.779	0.780	0.694
	AoC ₃ Reducing food waste at universities can substantially contribute to safeguarding the university ecosystem and the ecological environment.	0.822			
Personal norm (PN) (De Groot & Steg, 2009; Kim et al., 2022)	PN ₁ I have remorse for discarding food after meals at my university.	0.811			
	PN ₂ I would improve as an individual by stopping wasting food during my university meals.	0.814	0.835	0.836	0.668
	PN ₃ I perceive a moral duty to avoid food waste at my university.	0.821			
	PN ₄ Food waste contradicts my personal principles.	0.823			
Food waste reduction Intention (FWR) (Coşkun & Özbük, 2020; Kim et al., 2022)	FWR ₁ I intend to consume all my food without leaving any leftovers while at the university.	0.850	0.821	0.821	0.736

Note: CR = Composite Reliability; AVE = Average Variance Extracted.

Discriminant validity, which examines the degree of distinctiveness among different constructs, was operationalized via the HTMT ratio (Hair et al., 2019). HTMT values below 0.90 are generally considered acceptable, while a stricter threshold is below 0.85 (Hair et al., 2019; Henseler et al., 2015). All constructs in Table 2 meet the HTMT criterion for discriminant validity.

Table 2. Results of discriminative evaluation

Construct	AoC	AoR	FWR	PN
AoC	-	-	-	-
AoR	0.810	-	-	-
FWR	0.827	0.761	-	-
PN	0.765	0.703	0.684	-

Note: AoC = Awareness of consequence; AoR = Ascription of responsibility; PN = Personal norms; FWR = Food waste reduction intention; hyphens (-) denote diagonal cells (self-correlation of each construct, not calculated).

In accordance with the criteria established by (Hair et al., 2021), multicollinearity is evaluated using Variance Inflation Factor (VIF) values, which ideally should remain below 3.0 or 3.33. In this study, the observed inner-model VIF values spanned from 1.000 to 1.661, confirming the complete absence of multicollinearity issues.

4.3 Structural Framework

The results indicate that the R^2 coefficients for the dependent variables in the model, namely personal norms and intention to reduce food waste, are 0.437 and 0.520, respectively. This suggests moderate to relatively strong explanatory power of the independent factors, compared with benchmark values of 0.75, 0.50, and 0.25. The Q^2 values indicate the predictive relevance of the exogenous constructs and provide a comparative assessment of predictive accuracy. Predictive relevance is commonly categorized into three levels: 0.02, 0.15, and 0.35. The Q^2 values for personal norms and intention to reduce food waste are 0.467 and 0.475, respectively, demonstrating the sufficient relevance of the proposed model.

4.4 Hypothesis Testing

4.4.1 Direct and indirect effects

Table 3 shows that all hypotheses proposing direct associations among the constructs are supported, with $t > 2.57$ and $p < 0.05$. Awareness of the consequences of food waste positively influences personal norms ($\beta_{\text{AoC-PN}} = 0.418$, $p < 0.05$) and intention to reduce food waste ($\beta_{\text{AoC-FWR}} = 0.375$, $p < 0.05$), supporting H_1 and H_2 . In addition,

ascription of responsibility for food waste is positively associated with personal norms and intention to reduce food waste ($\beta_{\text{AoR-PN}} = 0.310$, $p < 0.05$; $\beta_{\text{AoR-FWR}} = 0.275$, $p < 0.05$), supporting H₃ and H₄. H₅ is also supported, indicating a positive association between personal norms and intention to reduce food waste ($\beta_{\text{PN-FWR}} = 0.177$, $p < 0.05$). Finally, H₆ and H₇ are supported, showing that awareness of consequence and ascription of responsibility indirectly influence students' intention to reduce food waste through personal norms ($\beta_{\text{AoC-PN-FWR}} = 0.449$, $p < 0.05$; $\beta_{\text{AoR-PN-FWR}} = 0.330$, $p < 0.05$).

Table 3. Results of hypothesis testing for direct and indirect effects

Direct Effects	β	p -Value	Findings
H ₁ . AoC → PN	0.418	***	Supported
H ₂ . AoC → FWR	0.375	***	Supported
H ₃ . AoR → PN	0.310	***	Supported
H ₄ . AoR → FWR	0.275	***	Supported
H ₅ . PN → FWR	0.177	***	Supported
Indirect effects	β	p -value	Findings
AoC → PN → FWR	0.074	**	
H ₆ . AoC → PN → FWR (total effect)	0.449	***	Supported
AoR → PN → FWR	0.055	***	
H ₇ . AoR → PN → FWR (total effect)	0.330	***	Supported

Note: *** $p < 0.001$; ** $p < 0.01$; AoC = Awareness of consequence; AoR = Ascription of responsibility; PN = Personal norms; FWR = Food waste reduction intention.

4.4.2 Multigroup analysis (MGA)

Regarding relationships within the study model, MGA is used to identify significant differences between the two groups of food buyers (OFD and IC). Before conducting MGA, it is necessary to assess the measurement invariance of composite models (MICOM) using a three-step procedure (Henseler et al., 2016). Tables 4 and 5 present the results of the three-step MICOM procedure. This is a prerequisite for evaluating significant differences between the two groups using MGA (Hair et al., 2014).

Table 4. Results of the invariance model assessment (Steps 1 & 2)

Constructs	Step 1	Step 2		Partial Measurement Invariance Established
		Compositional Original Correlation	Invariance Confidence Interval	
AoC	Yes	0.999	[0.999; 1.000]	Yes
AoR	Yes	0.999	[0.997; 0.999]	Yes
FWR	Yes	1.000	[0.999; 1.000]	Yes
PN	Yes	0.999	[0.999; 1.000]	Yes

Note: AoC = Awareness of consequence; AoR = Ascription of responsibility; PN = Personal norms; FWR = Food waste reduction intention.

Table 5. Results of the invariance model test (Step 3a & 3b)

Constructs	Step 3a			Step 3b			Full Measurement Invariance Established
	Equal Mean Assessment		Equal	Equal Mean Assessment		Equal	
	Mean Original Difference	Confidence Interval		Mean Original Difference	Confidence Interval		
AoC	0.094	[-0.159; 0.153]	Yes	0.052	[-0.369; 0.363]	Yes	Yes
AoR	0.057	[-0.168; 0.155]	Yes	-0.114	[-0.383; 0.368]	Yes	Yes
FWR	-0.117	[-0.168; 0.153]	Yes	0.127	[-0.376; 0.330]	Yes	Yes
PN	0.011	[-0.171; 0.161]	Yes	-0.091	[-0.427; 0.407]	Yes	Yes

Note: AoC = Awareness of consequence; AoR = Ascription of responsibility; PN = Personal norms; FWR = Food waste reduction intention.

The multi-group analysis results presented in Table 6 show that both the MGA and Welch-Satterthwaite tests yield p -values greater than 0.05 ($p_{\text{MGA}} = 0.056$; $p_{\text{Welch-Satterthwaite}} = 0.059$). However, the Parametric test produces a p -value less than 0.05 ($\beta_{\text{OFD-IC}} = 0.224$; $p_{\text{Parametric}} = 0.042$), thereby providing support for hypothesis H_{8a}. The effect of personal norms on the intention to reduce food waste was weaker among students using OFD than among those

dining in the canteen ($\beta_{\text{OFD-IC}} = -0.174$, $p_{\text{Parametric}} = 0.036$). Therefore, H_{8b} is supported. However, no significant differences were found in the relationship between ascription of responsibility and the intention to reduce food waste ($\beta_{\text{OFD-IC}} = -0.113$, $p_{\text{MGA}} = 0.227$; $p_{\text{Welch-Satterthwaite}} = 0.258$; $p_{\text{Parametric}} = 0.276$). Therefore, H_{8c} is not supported.

Table 6. Results of multi-group (online food delivery (OFD) and in-canteen dining (IC))

Hypothesis	Path Coefficient		Difference (OFD-IC)	p-Value 2-Tailed			Findings
	OFD	IC		MGA	Parametric test	Welch-Satterthwaite	
H_{8a} . AoC $\rightarrow$ FWR	0.523***	0.299***	0.224	0.056	0.042	0.059	Supported
H_{8b} . PN $\rightarrow$ FWR	0.064	0.238***	-0.174	0.049	0.036	0.046	Supported
H_{8c} . AoR $\rightarrow$ FWR	0.208***	0.321***	-0.113	0.277	0.258	0.276	Not Supported

Note: * $p < 0.05$, *** $p < 0.001$. AoC = Awareness of consequence; AoR = Ascription of responsibility; PN = Personal norms; FWR = Food waste reduction intention; MGA = Multi-Group Analysis.

5. Discussion and Theoretical Contribution

Previous studies on food waste reduction that draw on the NAT have largely focused on high-risk consumption contexts such as restaurants, tourism, and households. Addressing this gap, the present study examines food waste reduction in educational institutions, with a particular focus on universities, a setting prone to high levels of food waste among young consumers. The results, consistent with prior research (Rasool et al., 2021), indicate that when students recognize the environmental harm associated with excessive consumption, they are more likely to perceive collective responsibility for the adverse effects of food waste. This awareness encourages them to adopt moral standards to mitigate it. By investigating their subsequent impact on food waste behavior among university students, this study expands upon existing literature regarding the mechanisms that trigger awareness of consequence and the ascription of responsibility within personal norms. Empirically, direct relationships were established between awareness of consequence, ascription of responsibility, and students' intentions to reduce food waste. These findings suggest that these two factors not only activate personal norms but also shape students' intentions to minimize food waste.

At universities, students attend lectures on ethics, environmental concerns, and sustainable consumer behavior (Wu et al., 2019). They also participate in programs aligned with the Sustainable Development Goals that prioritize food waste reduction, including initiatives on food waste norms, environmental forums, and other activities. Such engagement explicitly encourages students' intention to minimize food waste (Kim et al., 2022). Furthermore, strengthened personal norms positively affect food waste reduction behavior, aligning with the findings of Kim et al. (2022) but contrasting with those of Mganga et al. (2021), who found that students' ethical attitudes did not significantly influence intentions to reduce food waste or related behaviors. University students constitute a distinct cohort with similar age and exposure to environmental education. Consequently, they may have a stronger understanding of the environmental system and more robust environmental attitudes (Schwartz & Boehnke, 2004) than the general population, reinforcing the role of ethical standards in shaping intentions to reduce food waste. Additionally, the relationship between awareness of consequence, ascription of responsibility, and the intention to minimize food waste is significantly mediated by personal norms. This suggests that the more clearly defined an individual's moral standards are, the more their awareness and attitudes toward environmental preservation translate into efforts to reduce food waste. This outcome aligns with Setiawan & Puspitasari (2023), who argued that evaluating responsibility is fundamental to the formation of personal norms and therefore serves an important mediating role.

Significantly, the unique academic value of this study lies in its exploration of how food purchasing modes—specifically OFD and IC—act as moderating variables within the proposed model. Compared with the IC group, students purchasing food via OFD reported greater awareness of the consequences and a stronger intention to reduce food waste. The results indicate that the OFD group values convenience, variety, and saving time and money, yet remains aware of the environmental ramifications of food waste, which in turn influences its intention to reduce it. However, the findings also show that students using OFD exhibit relationship between personal norms and intention to reduce food waste weaker than the IC group. The popularity of OFD among students can be attributed to its advantages, including convenience, shared delivery costs, discounts, and a wide selection of meal options. As a result, consumers may be influenced by these value-related factors when ordering through OFD, thereby weakening their understanding of waste and its implications. Lower food waste reduction behavior among OFD users is often linked to platform requirements such as a “minimum price”, which can lead consumers to purchase more than needed or coordinate orders with roommates to reach the minimum for free delivery. This can create difficulties in managing food surplus in line with ethical consumption (Wei et al., 2023). Conversely, even when students hold strong ethical values, consumption-stimulating policies and programs (such as discounts and gifts), combined with price sensitivity, may still constrain efforts to reduce food waste (Fraj-Andrés et al., 2022).

When dining at a canteen, consumers encounter several sensory cues, including aroma, visual appeals, portion size, and food dimensions, which help them judge an appropriate amount to eat and thereby minimize waste. This suggests that the economic incentives offered through OFD may override personal norms, leaving students less motivated to regulate food waste behavior.

6. Conclusion and Practical Contributions

OFD platform providers and restaurants, as principal stakeholders, are essential for tackling food waste. First, platforms could require providers to display more detailed information on application pages, such as portion sizes, ingredients, and clearer flavor descriptions, to increase consumer awareness and control over food choices. This can support platforms' efforts to reduce food waste by lowering the likelihood of food waste. In addition, because food quality and safety are important concerns for some consumers, displaying relevant certificates on OFD platforms could strengthen trust, encouraging consumers to complete purchases and to store leftovers for later use. Second, platform providers should avoid encouraging or incentivizing consumers to overpurchase, since this can increase food waste. While incentives and promotions can boost purchasing intention in the early stages of market development, reducing or removing them as the market matures can help curb unnecessary purchases beyond consumers' needs. Third, platforms can offer multiple portion options or provide suggested portion sizes to prompt consumers to select appropriate quantities and make decisions that reduce food waste.

Universities can strengthen public campaigns to improve students' knowledge of sustainable eating practices and use social media more effectively to increase awareness of food waste and its consequences. Such initiatives should target specific behaviors and help build university and community cultures in which participants recognize the impacts of food waste and understand how to reduce it. The more informed and aware individuals are, the more likely they are to adopt positive behavioral changes. Universities can also improve on-campus dining facilities (canteens and other food service outlets). When eating at the canteen, students tend to focus on their meals and are less likely to order more than they can consume, thereby reducing reliance on OFD and helping mitigate food waste on campus.

7. Limitations and Directions for Future Research

While providing important insights, the current research is subject to several shortcomings that illuminate potential pathways for future studies: (1) the utilization of convenience sampling, which potentially compromises the sample's representativeness; (2) The low student response rate may reflect some students' perception that they do not waste food, which reduced participation despite multiple questionnaire distributions; and (3) Data collection was limited to universities in Vietnam, potentially restricting the generalizability of the findings. Future studies could improve representativeness by using probability sampling and increasing sample sizes through incentives such as gifts or prizes to encourage participation. Future scholars could also examine cultural differences in food intake across countries.

Author Contributions

Conceptualization, H.T.T.H. and N.N.T.M.; data analysis, H.T.T.H. and N.N.T.M.; validation, H.T.T.H. and N.N.T.M.; writing—review & editing, H.T.T.H. and N.N.T.M.; software, N.N.T.M. and H.T.T.H.; formal analysis, N.N.T.M. and H.T.T.H.; methodology, N.N.T.M.; writing—original draft, N.N.T.M. and H.T.T.H. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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